

USE WIRESHARK TO ANALYZE PACKET CAPTURES AND DETECT A MAN-IN-THE- MIDDLE (MITM) ATTACK VIA ARP POISONING

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Digital Forensics – Project 2



INTRODUCTION

- Digital forensics plays a critical and key role in identifying, analyzing, and documenting cyber incidents.
- MITM attacks intercept and manipulate communication between two legitimate systems.
- ARP poisoning enables traffic redirection.
- Wireshark can be used to identify IoC's associated with ARP poisoning.
- Traffic analysis can be conducted using endpoint statistics and detailed packet inspection.
- MITM attack can be detected using ARP traffic patterns and anomalies in MAC address mappings.

OBJECTIVE

- Analyze provided packet capture *proj_part1.pcap* file using Wireshark.
- Identify ARP poisoning activity.
- Detect evidence of MITM attack.
- Document findings using forensic methods.

ADDRESS RESOLUTION PROTOCOL (ARP)

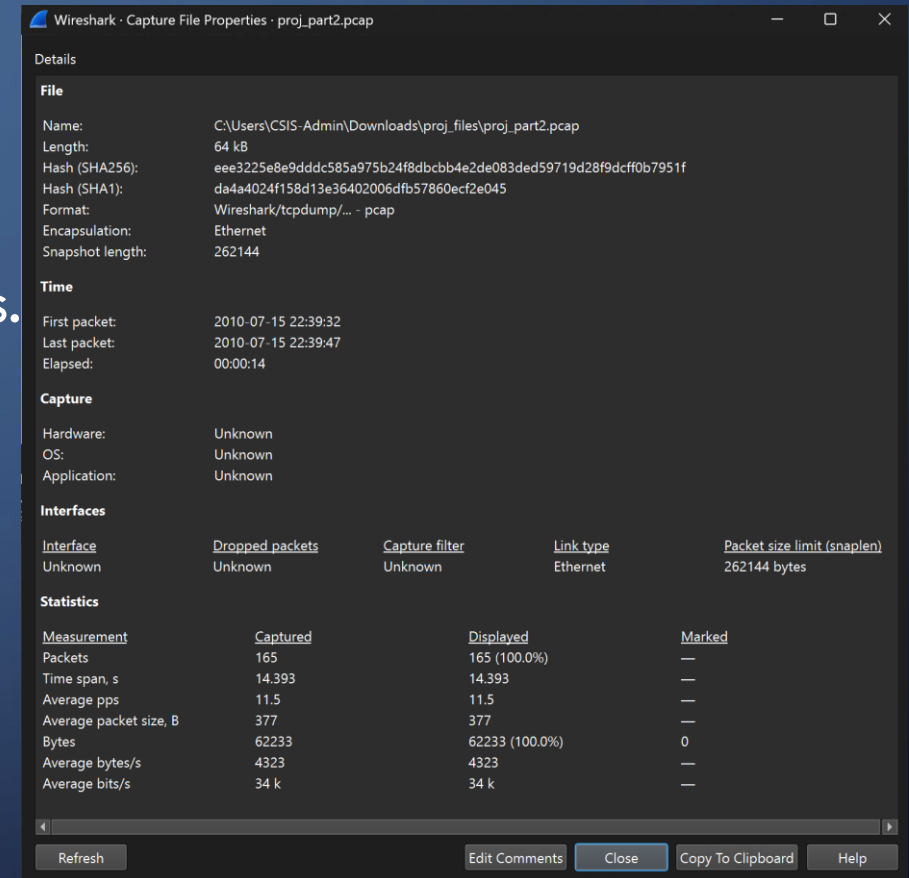
- Maps IP addresses to MAC addresses.
- Uses ARP request (broadcast).
- Uses ARP reply (unicast).
- Essential for local network communication.

MITM VIA ARP POISONING

- ARP uses request and response packets.
- Attacker sends fake ARP replies.
- Associates` attacker MAC with victim/router IP.
- Redirects traffic through attacker.
- Allows interception and modification of data.

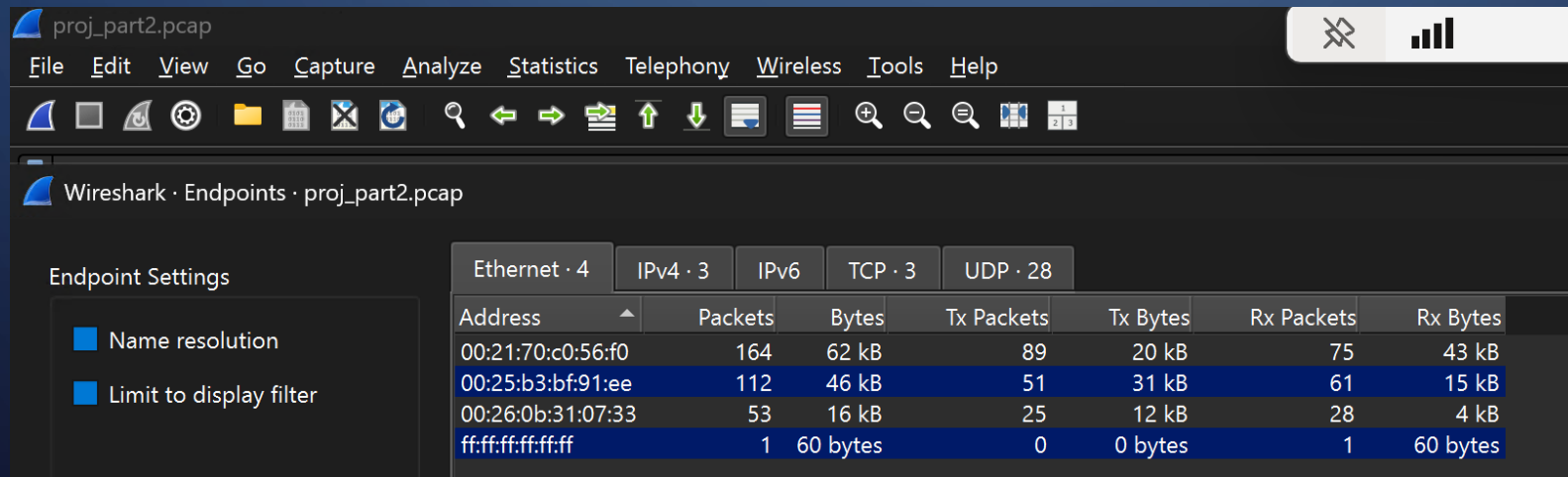
EVIDENCE VERIFICATION

- Verified file integrity using SHA256 hash.
- Ensures evidence is not altered.
- One of most important steps in digital forensics.
- Confirms reliability of analysis.



ENDPOINT ANALYSIS (ETHERNET)

- Identified victim MAC `00:21:70:c0:56:f0`
- Identified router MAC `00:26:0b:31:07:33`
- Detected attacker MAC `00:25:b3:bf:91:ee`
- Observed broadcast MAC address `ff:ff:ff:ff:ff:ff`



The screenshot shows the Wireshark interface with the 'Endpoints' pane open for the file 'proj_part2.pcap'. The 'Endpoint Settings' on the left show 'Name resolution' and 'Limit to display filter' are both checked. The main pane displays a table of statistics for various network protocols. The 'Ethernet' tab is selected, showing a list of MAC addresses and their corresponding packet and byte counts for transmission and reception.

Wireshark · Endpoints · proj_part2.pcap							
Endpoint Settings							
☒ Name resolution							
☒ Limit to display filter							
Ethernet · 4 IPv4 · 3 IPv6 TCP · 3 UDP · 28							
Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	
00:21:70:c0:56:f0	164	62 kB	89	20 kB	75	43 kB	
00:25:b3:bf:91:ee	112	46 kB	51	31 kB	61	15 kB	
00:26:0b:31:07:33	53	16 kB	25	12 kB	28	4 kB	
ff:ff:ff:ff:ff:ff	1	60 bytes	0	0 bytes	1	60 bytes	

ENDPOINT ANALYSIS (IPV4)

- Identified victim IPv4 172.16.0.107
- Identified IPv4 addresses 12.153.20.41, and 74.125.95.147

Wireshark · Endpoints · proj_part2.pcap

Endpoint Settings

☐ Name resolution

☒ Limit to display filter

Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country
12.153.20.41	54	10 kB	27	8 kB	27	2 kB	
74.125.95.147	107	52 kB	46	34 kB	61	17 kB	
172.16.0.107	161	62 kB	88	20 kB	73	43 kB	

ENDPOINT ANALYSIS (TCP)

- Displayed communication between the victim (172.16.0.107) and an external IP address (74.125.95.147) over port 80.
- Multiple packets and bytes indicate ongoing data exchange.

Wireshark · Endpoints · proj_part2.pcap

Endpoint Settings

☐ Name resolution

☒ Limit to display filter

Ethernet · 4

IPv4 · 3

IPv6

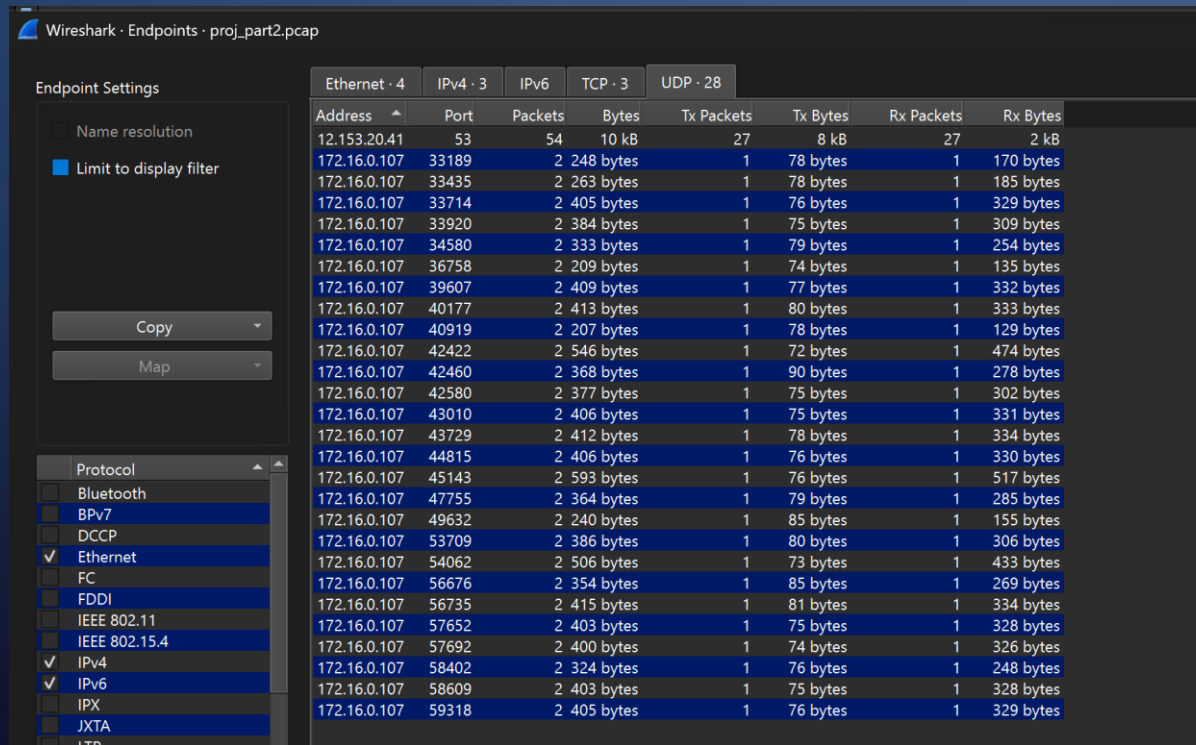
TCP · 3

UDP · 28

Address	Port	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
74.125.95.147	80	107	52 kB	46	34 kB	61	17 kB
172.16.0.107	45691	72	39 kB	40	10 kB	32	29 kB
172.16.0.107	45692	35	13 kB	21	8 kB	14	5 kB

ENDPOINT ANALYSIS (UDP)

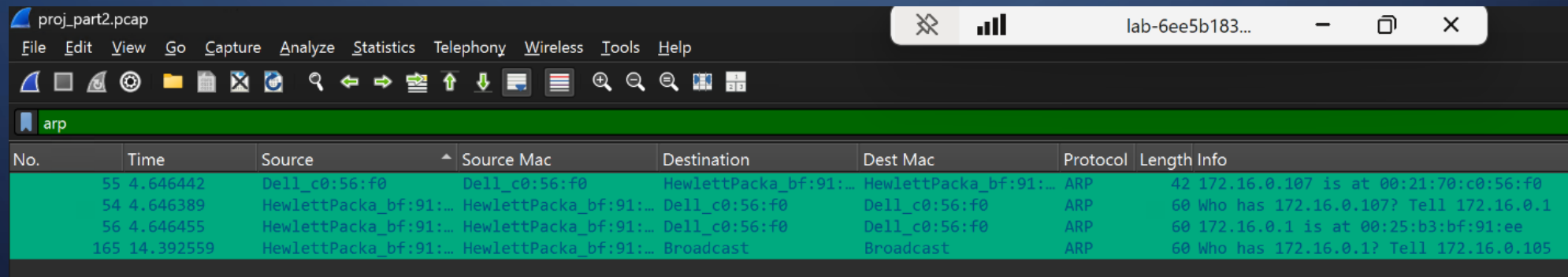
- Multiple short communications primarily involving the victim (172.16.0.107) using various ephemeral ports.
- DNS traffic over UDP (port 53) was observed.



Address	Port	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
12.153.20.41	53	54	10 kB	27	8 kB	27	2 kB
172.16.0.107	33189	2	248 bytes	1	78 bytes	1	170 bytes
172.16.0.107	33435	2	263 bytes	1	78 bytes	1	185 bytes
172.16.0.107	33714	2	405 bytes	1	76 bytes	1	329 bytes
172.16.0.107	33920	2	384 bytes	1	75 bytes	1	309 bytes
172.16.0.107	34580	2	333 bytes	1	79 bytes	1	254 bytes
172.16.0.107	36758	2	209 bytes	1	74 bytes	1	135 bytes
172.16.0.107	39607	2	409 bytes	1	77 bytes	1	332 bytes
172.16.0.107	40177	2	413 bytes	1	80 bytes	1	333 bytes
172.16.0.107	40919	2	207 bytes	1	78 bytes	1	129 bytes
172.16.0.107	42422	2	546 bytes	1	72 bytes	1	474 bytes
172.16.0.107	42460	2	368 bytes	1	90 bytes	1	278 bytes
172.16.0.107	42580	2	377 bytes	1	75 bytes	1	302 bytes
172.16.0.107	43010	2	406 bytes	1	75 bytes	1	331 bytes
172.16.0.107	43729	2	412 bytes	1	78 bytes	1	334 bytes
172.16.0.107	44815	2	406 bytes	1	76 bytes	1	330 bytes
172.16.0.107	45143	2	593 bytes	1	76 bytes	1	517 bytes
172.16.0.107	47755	2	364 bytes	1	79 bytes	1	285 bytes
172.16.0.107	49632	2	240 bytes	1	85 bytes	1	155 bytes
172.16.0.107	53709	2	386 bytes	1	80 bytes	1	306 bytes
172.16.0.107	54062	2	506 bytes	1	73 bytes	1	433 bytes
172.16.0.107	56676	2	354 bytes	1	85 bytes	1	269 bytes
172.16.0.107	56735	2	415 bytes	1	81 bytes	1	334 bytes
172.16.0.107	57652	2	403 bytes	1	75 bytes	1	328 bytes
172.16.0.107	57692	2	400 bytes	1	74 bytes	1	326 bytes
172.16.0.107	58402	2	324 bytes	1	76 bytes	1	248 bytes
172.16.0.107	58609	2	403 bytes	1	75 bytes	1	328 bytes
172.16.0.107	59318	2	405 bytes	1	76 bytes	1	329 bytes

ARP IDENTIFICATION

- Used Wireshark coloring rules.
- Applied display filter arp
- Isolated ARP request and response packets
- Enabled focused ARP traffic analysis

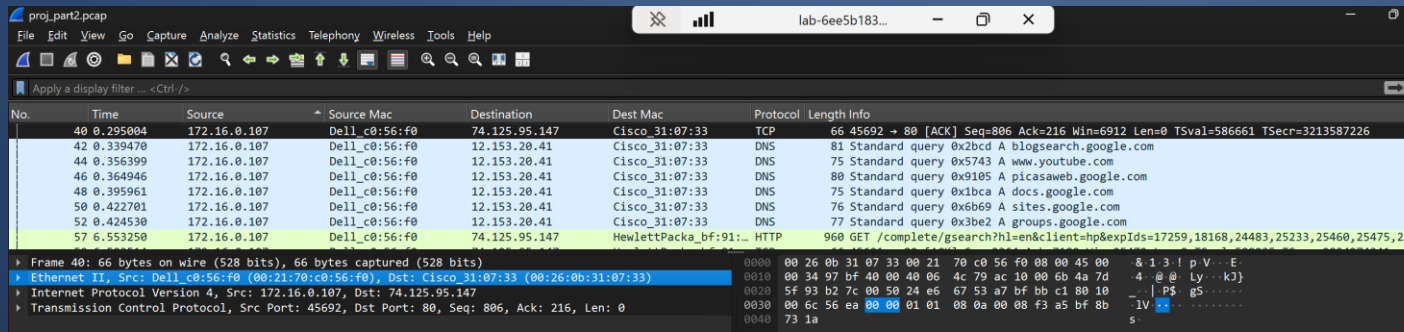


The image shows a Wireshark window titled 'proj_part2.pcap'. The display filter is set to 'arp'. The packet list shows four ARP-related packets. The packet details pane shows the selected packet (No. 165) with fields for Ethernet II, Internet Protocol Version 4, and ARP. The ARP section shows 'Who has 172.16.0.1? Tell 172.16.0.105'.

No.	Time	Source	Source Mac	Destination	Dest Mac	Protocol	Length	Info
55	4.646442	Dell_c0:56:f0	Dell_c0:56:f0	HewlettPacka_bf:91:...	HewlettPacka_bf:91:...	ARP	42	172.16.0.107 is at 00:21:70:c0:56:f0
54	4.646389	HewlettPacka_bf:91:...	HewlettPacka_bf:91:...	Dell_c0:56:f0	Dell_c0:56:f0	ARP	60	Who has 172.16.0.107? Tell 172.16.0.1
56	4.646455	HewlettPacka_bf:91:...	HewlettPacka_bf:91:...	Dell_c0:56:f0	Dell_c0:56:f0	ARP	60	172.16.0.1 is at 00:25:b3:bf:91:ee
165	14.392559	HewlettPacka_bf:91:...	HewlettPacka_bf:91:...	Broadcast	Broadcast	ARP	60	Who has 172.16.0.1? Tell 172.16.0.105

PACKET ANALYSIS (PACKET 40 AND 57)

- Packet 40 - Normal communication (victim to router)

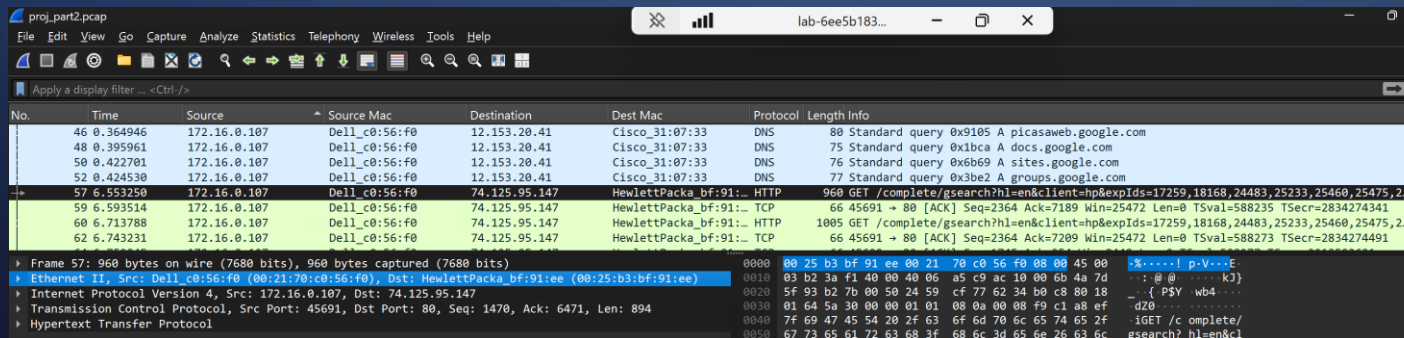


The screenshot shows a Wireshark packet capture of packet 40. The packet list on the left shows packet 40 as an Ethernet II frame from Dell_c0:56:f0 to Cisco_31:07:33. The packet details pane shows the Ethernet II header, Internet Protocol Version 4 header, and Transmission Control Protocol header. The packet bytes pane shows the raw data of the packet.

No.	Time	Source	Source Mac	Destination	Dest Mac	Protocol	Length	Info
40	0.295004	172.16.0.107	Dell_c0:56:f0	74.125.95.147	Cisco_31:07:33	TCP	66	45692 → 80 [ACK] Seq=806 Ack=216 Win=6912 Len=0 TSval=586661 TSecr=3213587226
42	0.339470	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	81	Standard query 0x2bcd A blogsearch.google.com
44	0.356399	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	75	Standard query 0x5743 A www.youtube.com
46	0.364946	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	80	Standard query 0x9105 A picasaweb.google.com
48	0.395961	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	75	Standard query 0x1bca A docs.google.com
50	0.422701	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	76	Standard query 0x6b69 A sites.google.com
52	0.424530	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	77	Standard query 0x3be2 A groups.google.com
57	6.553250	172.16.0.107	Dell_c0:56:f0	74.125.95.147	HewlettPacka_bf:91:...	HTTP	960	GET /complete/gsearch?hl=en&client=hp&expids=17259,18168,24483,25233,25460,25475,2...

Frame 40: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0
Ethernet II, Src: Dell_c0:56:f0 (00:21:70:c0:56:f0), Dst: Cisco_31:07:33 (00:26:0b:31:07:33)
Internet Protocol Version 4, Src: 172.16.0.107, Dst: 74.125.95.147
Transmission Control Protocol, Src Port: 45692, Dst Port: 80, Seq: 806, Ack: 216, Len: 0

- Packet 57 - MAC address changes to attacker



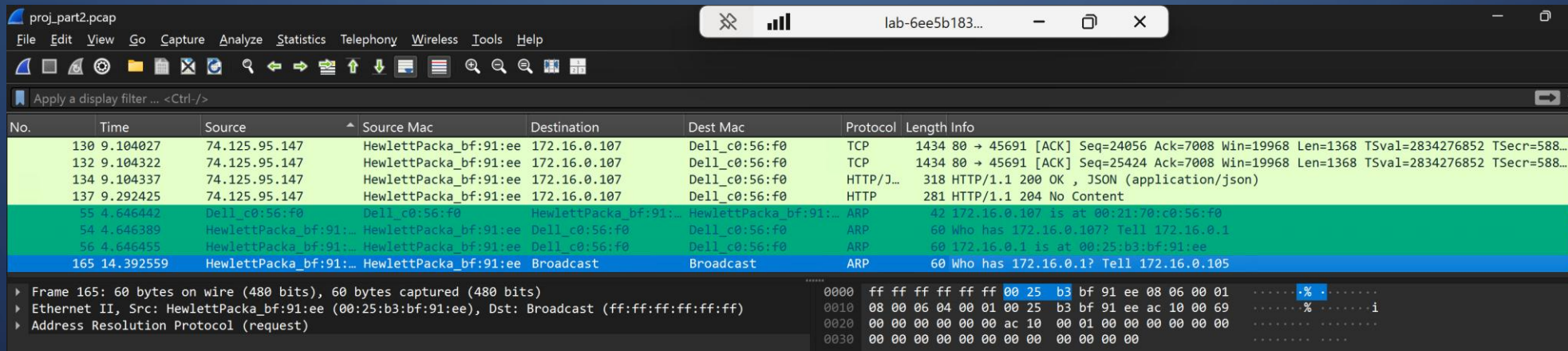
The screenshot shows a Wireshark packet capture of packet 57. The packet list on the left shows packet 57 as an Ethernet II frame from Dell_c0:56:f0 to HewlettPacka_bf:91:ee. The packet details pane shows the Ethernet II header, Internet Protocol Version 4 header, and Hypertext Transfer Protocol header. The packet bytes pane shows the raw data of the packet.

No.	Time	Source	Source Mac	Destination	Dest Mac	Protocol	Length	Info
46	0.364946	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	80	Standard query 0x9105 A picasaweb.google.com
48	0.395961	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	75	Standard query 0x1bca A docs.google.com
50	0.422701	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	76	Standard query 0x6b69 A sites.google.com
52	0.424530	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	77	Standard query 0x3be2 A groups.google.com
57	6.553250	172.16.0.107	Dell_c0:56:f0	74.125.95.147	HewlettPacka_bf:91:ee	HTTP	960	GET /complete/gsearch?hl=en&client=hp&expids=17259,18168,24483,25233,25460,25475,2...
59	6.593514	172.16.0.107	Dell_c0:56:f0	74.125.95.147	HewlettPacka_bf:91:ee	TCP	66	45691 → 80 [ACK] Seq=2364 Ack=7189 Win=25472 Len=0 TSval=588235 TSecr=2834274341
60	6.713788	172.16.0.107	Dell_c0:56:f0	74.125.95.147	HewlettPacka_bf:91:ee	HTTP	1005	GET /complete/gsearch?hl=en&client=hp&expids=17259,18168,24483,25233,25460,25475,2...
62	6.743231	172.16.0.107	Dell_c0:56:f0	74.125.95.147	HewlettPacka_bf:91:ee	TCP	66	45691 → 80 [ACK] Seq=2364 Ack=7209 Win=25472 Len=0 TSval=588273 TSecr=2834274491

Frame 57: 960 bytes on wire (7680 bits), 960 bytes captured (7680 bits) on interface 0
Ethernet II, Src: Dell_c0:56:f0 (00:21:70:c0:56:f0), Dst: HewlettPacka_bf:91:ee (00:25:b3:bf:91:ee)
Internet Protocol Version 4, Src: 172.16.0.107, Dst: 74.125.95.147
Transmission Control Protocol, Src Port: 45691, Dst Port: 80, Seq: 1470, Ack: 6471, Len: 894
Hypertext Transfer Protocol

PACKET ANALYSIS (PACKET 165)

- Packet 165 - ARP request broadcast by attacker.
- Indicates traffic redirection and manipulation.



The image shows a Wireshark packet capture analysis of a file named 'proj_part2.pcap'. The interface includes a menu bar, a toolbar, and a packet list pane. The packet list pane shows several packets, with packet 165 highlighted in blue. Packet 165 is an ARP request broadcast from source 'HewlettPacka_bf:91:ee' to destination 'Broadcast'. The packet details pane shows the frame structure: Ethernet II, Src: HewlettPacka_bf:91:ee (00:25:b3:bf:91:ee), Dst: Broadcast (ff:ff:ff:ff:ff:ff), and Address Resolution Protocol (request). The packet bytes pane shows the raw data of the packet, with the ARP request structure visible.

No.	Time	Source	Source Mac	Destination	Dest Mac	Protocol	Length	Info
130	9.104027	74.125.95.147	HewlettPacka_bf:91:ee	172.16.0.107	Dell_c0:56:f0	TCP	1434	80 → 45691 [ACK] Seq=24056 Ack=7008 Win=19968 Len=1368 TSval=2834276852 TSecr=588...
132	9.104322	74.125.95.147	HewlettPacka_bf:91:ee	172.16.0.107	Dell_c0:56:f0	TCP	1434	80 → 45691 [ACK] Seq=25424 Ack=7008 Win=19968 Len=1368 TSval=2834276852 TSecr=588...
134	9.104337	74.125.95.147	HewlettPacka_bf:91:ee	172.16.0.107	Dell_c0:56:f0	HTTP/J...	318	HTTP/1.1 200 OK , JSON (application/json)
137	9.292425	74.125.95.147	HewlettPacka_bf:91:ee	172.16.0.107	Dell_c0:56:f0	HTTP	281	HTTP/1.1 204 No Content
55	4.646442	Dell_c0:56:f0	Dell_c0:56:f0	HewlettPacka_bf:91:ee	HewlettPacka_bf:91:ee	ARP	42	172.16.0.107 is at 00:21:70:c0:56:f0
54	4.646389	HewlettPacka_bf:91:ee	HewlettPacka_bf:91:ee	Dell_c0:56:f0	Dell_c0:56:f0	ARP	60	Who has 172.16.0.107? Tell 172.16.0.1
56	4.646455	HewlettPacka_bf:91:ee	HewlettPacka_bf:91:ee	Dell_c0:56:f0	Dell_c0:56:f0	ARP	60	172.16.0.1 is at 00:25:b3:bf:91:ee
165	14.392559	HewlettPacka_bf:91:ee	HewlettPacka_bf:91:ee	Broadcast	Broadcast	ARP	60	Who has 172.16.0.1? Tell 172.16.0.105

Frame 165: 60 bytes on wire (480 bits), 60 bytes captured (480 bits)
Ethernet II, Src: HewlettPacka_bf:91:ee (00:25:b3:bf:91:ee), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Address Resolution Protocol (request)

0000 ff ff ff ff ff ff 00 25 b3 bf 91 ee 08 06 00 01%.....
0010 08 00 06 04 00 01 00 25 b3 bf 91 ee ac 10 00 69%.....i
0020 00 00 00 00 00 00 ac 10 00 01 00 00 00 00 00
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00

FINDINGS

- MAC spoofing detected.
- ARP poisoning confirmed.
- Traffic redirected through attacker.
- Persistent malicious ARP activity observed.

CONCLUSION

- Hash validation ensures integrity and reliability of the data analyzed.
- Confirms the presence of a Man-in-the-Middle (MITM) attack using ARP poisoning.
- Victim device communicated with external systems through the router indicates the Normal traffic.
- Attacker positioned between victim and router.
- ARP request is being broadcast by the attacker MAC address to the network.
- Highlights importance of network monitoring and forensic analysis in detecting and responding to security incidents.

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THANK YOU!